# Exercise Sheet 3

## Keep the Peak, the Generative Route and Naive Bayes

Probabilistic Machine Learning

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# 3.A Keep the Peak

# Exercise 3.A.1: Peak, Mean and a Prior's Peak

Returning users, $\theta = P(\text{user returns})$, prior $\text{Beta}(a, b)$, $h$ of $n$ users return.

1. Maximise $\log p(D \mid \theta) + \log p(\theta)$ and show $\hat{\theta}_{\text{MAP}} = \frac{h+a-1}{n+a+b-2}$.
2. **0 of 4** users return. Give the MLE, the MAP under $\text{Beta}(2, 2)$ and the posterior mean under $\text{Beta}(2, 2)$. Which one would you report to a product manager, and why?
3. The posterior mean under a flat prior is $\frac{h+1}{n+2}$. Under which prior is the **MAP** equal to it?
4. After 3 of 3 with a flat prior the posterior is $\text{Beta}(4, 1)$, with density $4\theta^3$. Compute its central 95% interval.

# 3.B The Generative Route

# Exercise 3.B.1: Counting Parameters

A fraud classifier (two classes) uses four attributes: **payment** (card, invoice, wallet), **country** (domestic, EU, other), **new device** (yes/no) and **night** (yes/no).

1. How many free numbers does the full generative model need — the class prior and one joint $p(x \mid y)$ per class?
2. How many does naive Bayes need?
3. For $p$ yes/no attributes and $K$ classes, give both counts as formulas.
4. From which $p$ on does the full joint need more than a million numbers *per class*?

# Exercise 3.B.2: Fitting Is Counting

An attribute takes $k$ values. Within one class we observed it $n$ times, value $v$ exactly $n_v$ times ($\sum_v n_v = n$). Model it as categorical with parameters $\theta_1, \ldots, \theta_k \geq 0, \sum_v \theta_v = 1$.

1. Write down the log-likelihood $\ell(\theta)$ of the $n$ i.i.d. observations.
2. For $k = 2$, substitute $\theta_2 = 1 - \theta_1$ and show that the MLE is $\hat{\theta}_1 = n_1/n$ — the $h/n$ of Lecture 03.
3. For general $k$, maximise $\ell$ under the constraint $\sum_v \theta_v = 1$ (Lagrange multiplier) and show $\hat{\theta}_v = n_v/n$.
4. Why can naive Bayes fit every attribute's table separately, while a full joint $p(x \mid y)$ could not be fitted attribute by attribute?

# 3.C Naive Bayes by Hand

# Exercise 3.C.1: Twelve Transactions

Flags: new device $D$, foreign IP $F$, night $N$, high amount $H$ (1 = yes).

| # | $D$ | $F$ | $N$ | $H$ | fraud? | # | $D$ | $F$ | $N$ | $H$ | fraud? |
|---|---|---|---|---|---|---|---|---|---|---|---|
| 1 | 1 | 1 | 1 | 1 | yes | 7 | 1 | 0 | 0 | 0 | no |
| 2 | 1 | 1 | 0 | 1 | yes | 8 | 0 | 0 | 0 | 0 | no |
| 3 | 1 | 0 | 1 | 0 | yes | 9 | 0 | 1 | 0 | 0 | no |
| 4 | 0 | 1 | 1 | 1 | yes | 10 | 0 | 0 | 0 | 0 | no |
| 5 | 0 | 0 | 0 | 0 | no | 11 | 1 | 0 | 1 | 0 | no |
| 6 | 0 | 0 | 1 | 0 | no | 12 | 0 | 0 | 0 | 0 | no |

1. Estimate $P(\text{fraud})$ and $P(\text{flag} = 1 \mid \text{class})$ for all four flags by counting.
2. Compute $P(\text{fraud} \mid x)$ for $x_1 = (D, F, N, H) = (1, 0, 0, 0)$.
3. Do the same for $x_2 = (0, 0, 0, 1)$. What goes wrong, and why?
4. Repeat 2 and 3 with Laplace smoothing on the flags (keep the prior unsmoothed). Which result changes a lot, which only a little?

# Exercise 3.C.2: Smoothing Is a Prior

A yes/no flag was seen $h$ times among $n$ frauds.

1. With a $\mathrm{Beta}(1,1)$ prior on $\theta = P(\mathrm{flag} \mid \mathrm{fraud})$, what is the posterior (Lecture 02)? Show that its **mean** is Laplace's $(h+1)/(n+2)$.
2. What is the posterior's **peak** for $h = 0$? What does that say about "keep the peak"?
3. "Add-$\alpha$" smoothing uses $(h+\alpha)/(n+2\alpha)$. Which prior belongs to it? Compute the estimate for $h = 0, n = 100$ and $\alpha = 0.5, 1, 10$.
4. A merchant category with $k = 10$ values; one category never occurred in 100 frauds. What does Laplace smoothing give, and which prior is behind it?